Testing

Invest in tests proportional to where data actually enters your control

Test investment by layer

Heaviest where data enters your control, lighter downstream

Staging	Heaviest, first layer you control
Intermediate	Selective, only join/agg risk
Marts	Business-logic, use-case specific

Singular test

A business rule too specific for a generic test to express

```
select order_id from {{ ref('fct_orders') }}
where order_total < 0
```

Generic tests

Reusable, parameterized, the cheap default for a reason

```
columns:
  - name: order_id
    tests: [not_null, unique]
  - name: status
    tests:
      - accepted_values:
          values: ['open', 'closed']
```

Unit tests

Correctness of the logic itself, tested against fixtures

```
unit_tests:
  - name: test_valid_email
    model: dim_customers
    given:
      - input: ref('stg_customers')
        rows:
          - {email: a@b.com, valid: true}
    expect:
      rows:
        - {is_valid_email_address: true}
```

```
dbt test --select "test_type:unit"
```

Run only these

Run unit tests locally

Skip warehouse compute entirely while iterating

```
export DBT_ENGINE_EXPERIMENTAL_LOCAL_UNIT_TESTS=true
```

Opt in

```
config: {compute: local}
```

Per test, runs on DuckDB

```
Snowflake or BigQuery only
```

Other adapters: remote

Package stack

Three different jobs, not three ways to do one thing

dbt_utils	Primary key testing
dbt_expectations	Richer assertions, seeds
elementary	Anomaly detection, auto tests

Route failures to the owner

A shared channel has no clock running on it

meta.owner tag	Resolves to the right channel
Shared #data-alerts channel	Nobody owns the clock

Test cost tiers

Spend compute where the risk actually is

not_null, unique	Every build
Cross-table reconciliation	Scheduled, prod only

Coverage check

Catches an untested model, human or agent-written alike

```
select m.unique_id
from {{ info_schema('models') }} as m
left join {{ info_schema('tests') }} as t
  on t.model_unique_id = m.unique_id
where m.access = 'public' and t.unique_id is null
```

dbt check	Runs it